
EN 505P: Energy Systems Laboratory

Course Code	EN 505P
Course Name	Energy Systems Laboratory
L-T-P-C	0-0-4-2
Prerequisites	None
Distribution	Core for M.Tech. Energy Engineering
Approval	10th Senate (Jun 2016)

Laboratory Modules

- Study of properties of fuel oils & biomass, calorific value of Biomass samples
- Testing of Gasifier: Effect of feedstock on heat output and pollutants
- Energy conservation study of IC Engine
- Fabrication of photovoltaic solar cell
- Performance assessment of PV modules: I-V Characteristics and Efficiency of a Solar PV cell
- Measurement of Solar Radiation and study of efficiency of a Flat Plate Solar Collector
- Piezo-electric energy harvesting: comparison of conversion efficiency of different materials and effect of vibration frequency on conversion efficiency
- Electrical machines characteristics
- Pollutant level measurement
- Boiler, Pump & motor efficiency testing
- Characterization of Refrigeration system
- Study of fuel cells and measurement of their efficiency
- Implementation of a solar powered pump
- Design and implementation of a solar powered home with 24/7 availability of electricity
- Visit of Hydro-electric power plant
- Visit of Thermal power plant